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PERSPECTIVE

Toward bridging the gap between machine intelligence and machine wisdom: Dilemmas and conjectures

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In recent years, artificial intelligence (AI) has achieved tremendous development, akin to a significant leap, similar to progressing from 1 to 100. However, a significant gap still exists between current machine intelligence and human wisdom: machine intelligence is constrained to post hoc inference based on existing data, lacking the ability for genuine exploratory innovation and possessing no prospective reasoning inherent to human wisdom. Drawing inspiration from human wisdom, this article presents conjectures for overcoming the four dilemmas faced by machine intelligence: neglect of silicon-based cognition, lack of artistry, pitfall of perfectionism, and obsession with uniformity. These conjectures aim to propel machine intelligence toward machine wisdom, achieving a great leap from 1 to i.

INTRODUCTION

The evolution of machine intelligence has followed a non-linear trajectory, characterized by significant setbacks and periods of stagnation, often referred to as artificial intelligence (AI) winters. These challenges have complicated the progression from foundational research to advanced applications. However, recent breakthroughs in deep reinforcement learning[1] and large language models have substantially advanced the field, enabling the integration of AI into diverse areas of human life[2] and, in some cases, exceeding human-level performance.[3]

However, the evolution of machine intelligence faces inherent limitations, as a significant gap remains between AI and human wisdom. As highlighted by Oxford researchers,[4] AI relies predominantly on pattern recognition and data-driven learning, restricting it to replicating existing knowledge rather than generating novel insights. Unlike human cognition, which advances through hypothesis formulation and empirical verification, machine intelligence lacks imagination and is confined to post hoc inferences based on available data, failing to achieve prospective reasoning.

The critical question remains: how can we evolve machine intelligence into machine wisdom? By examining the genesis and development of human cognition, as Figure 1 shows, we identify four principal challenges that differentiate contemporary machine intelligence from human wisdom: the neglect of silicon-based cognition, the lack of artistry, the trap of perfection, and the obsession with uniformity. Addressing these limitations is essential for advancing machine intelligence to its next evolutionary phase, ultimately achieving a transformative leap.

BUILDING INTERPRETABLE META-KNOWLEDGE FOR SILICON-BASED COGNITION

The neglect of silicon-based cognition refers to the oversight in developing comprehensible knowledge systems for silicon-based entities such as machine intelligence. Unlike carbon-based humans, who perceive the world in three dimensions and can develop knowledge systems by drawing reasonable

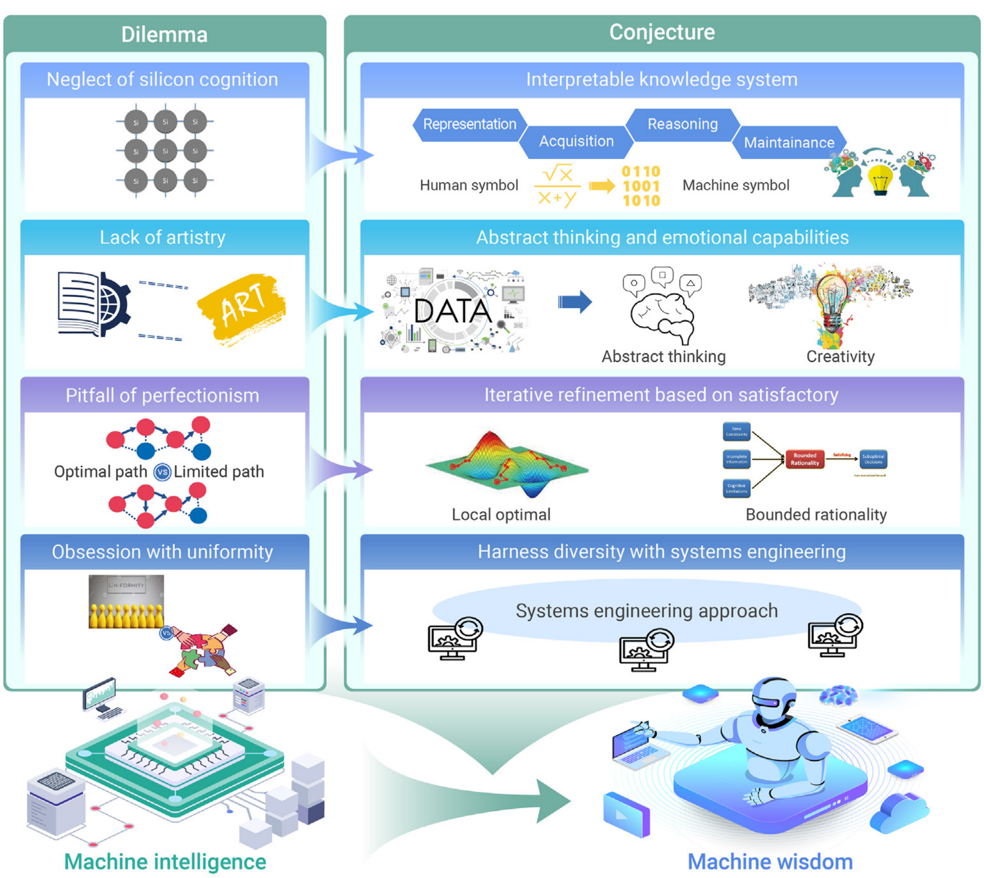

Figure 1. Investigate conjectures and paths for AI to break through bottlenecks and reach machine wisdom Four main dilemmas include the neglect of silicon-based cognition, the lack of artistry, the pitfall of perfectionism, and the obsession with uniformity. The corresponding conjecture and path for each dilemma is an interpretable knowledge system, abstract thinking and emotional capabilities, iterative refinement based on satisfactory, and harness diversity with systems engineering.

inferences from limited data, machine intelligence inherently relies heavily on learning from extensive datasets and lacks the ability to autonomously generate new hypotheses.[5]

Human wisdom, developed through lifelong learning and knowledge accumulation, produces a stable and widely accepted knowledge system. This system integrates diverse fields, including natural sciences, social sciences, and humanities, forming the foundation for human understanding. A well-structured and coherent knowledge system allows people to systematically organize information, enhancing their ability to comprehend and apply knowledge effectively.

Addressing the oversight in silicon cognition requires developing a meta-knowledge system tailored for machine intelligence. Andrew Ng advocates for a pragmatic approach to this development, emphasizing the pivotal role of data,[6] while Kai-Fu Lee underscores the necessity of integrating interdisciplinary knowledge and highlights the importance of ethics and societal impacts to mitigate potential risks. The meta-knowledge system is structured into four core components: knowledge representation, acquisition, reasoning, and maintenance. The knowledge representation component utilizes advanced technologies, such as knowledge graphs and neural networks, to efficiently encode and structure knowledge. Knowledge acquisition employs methodologies such as web crawling, text analysis, and multimodal integration to build a robust knowledge repository aligned with the chosen representation frameworks. Knowledge reasoning applies rule-based, model-based, and probabilistic approaches to refine existing knowledge and infer new insights. Lastly, knowledge maintenance encompasses validation, verification, and version control mechanisms to ensure the accuracy, consistency, and timeliness of the knowledge repository.

CULTIVATING ABSTRACT THINKING AND EMOTIONAL ARTISTRY

The absence of artistic literacy in silicon-based entities arises from machine intelligence's dependence on rule-based logic and data-driven processes, which are fundamentally incompatible with the abstract thinking essential to art. Margaret Boden contends that while machines can simulate aesthetic processes by analyzing elements such as color and composition, they lack genuine aesthetic appreciation or emotional depth due to their absence of consciousness and intrinsic emotions.[7] For example, DALL-E 2 generates visually impressive images, aiding brainstorming and design, but it cannot replicate the emotional resonance and contextual nuance derived from human artists' personal experiences and intentional storytelling. This limitation is further compounded by ethical concerns, as machine models often rely on copyrighted datasets without consent, potentially devaluing human creativity and livelihoods. As a result, machine-generated content often lacks true originality, merely rehashing existing data without innovation. Furthermore, machines cannot deeply empathize with human emotions, relying only on emotion-related vocabulary, and demonstrate limited cross-domain adaptability, constraining their utility across varied contexts.

Human wisdom harmoniously integrates artistic beauty and scientific rationality through the synergy of abstract and logical thinking. Abstract thinking extracts general principles from specific instances, driving innovation and theoretical advancement, while logical thinking employs rule-based reasoning and analysis to validate conclusions. This integration requires both creativity and rigorous argumentation in problem solving, resulting in insights that are both profound and practically valuable. Together, these elements embody the depth and richness of human wisdom.

To address the lack of art in machines, integrating logic and abstraction is essential. Xuesen Qian distinguished intelligence into quantitative (logical) and qualitative (abstract) thinking, asserting that both scientific and artistic endeavors require their synthesis. Similarly, Herbert Simon, a pioneer in AI, posited that machine creativity could be achieved by combining these modes, enabling innovative thinking.[8] Two promising approaches to imbue machines with artistic understanding are the hypothesis-driven method and the integration of Eastern and Western cognitive paradigms. The hypothesis-driven approach involves the generation of multiple artistic solutions based on characteristics such as color, composition, and rhythm, which are then evaluated against theoretical and stylistic criteria. Meanwhile, blending Eastern thinking—characterized by holistic, intuitive perspectives—with Western thinking—grounded in precision and logical analysis—enhances machine creativity. This synthesis leverages the strengths of both paradigms, fostering deeper artistic comprehension and versatility in machine intelligence.

REFINING ITERATIVELY TO AVOID PERFECTIONISM

The pitfall of perfectionism points out that machine intelligence often falls into the trap of pursuing optimal solutions, leading to issues such as high computational complexity and resource consumption. In contrast, human problem solving also seeks optimality but prioritizes satisfactory solutions within specific constraints. These solutions are sufficiently good without being perfect, offering greater efficiency, faster execution, and increased flexibility.

Human wisdom entails making locally optimal decisions within the framework of bounded rationality, as proposed by Herbert Simon, rather than pursuing global optimality through unlimited rationality. Bounded rationality posits that individuals, constrained by limited time and cognitive resources in complex environments, prioritize satisfactory solutions over ideal ones. This approach underscores the pragmatic balance between efficiency and feasibility, emphasizing reasonably effective outcomes in real-world contexts over theoretical perfection.

To avoid the trap of perfection, it is essential to adopt locally optimal solutions based on bounded rationality. Perfection is unattainable, and developing solutions without practical considerations is inadvisable. Steve Jobs prioritized user experience in product design and innovation over technical perfection, resulting in the highly acclaimed Apple product line. Similarly, Wu Jun, a former senior researcher at Google, argues that machine intelligence should integrate with practical applications to find locally optimal solutions that address current problems and create value rather than blindly pursuing advanced technologies and perfect solutions. An effective strategy involves optimizing domain-specific machines to reduce the search space and enhance efficiency, thereby quickly obtaining feasible solutions even if they are not optimal. It also requires continuously adjusting solutions based on user and market feedback to better meet actual needs and increasing the ability to handle uncertainty and ambiguity to ensure the solutions are both flexible and robust.

HARNESSING DIVERSITY WITH SYSTEMS ENGINEERING

The obsession with uniformity constitutes another obstacle in the development of machine intelligence. In the pursuit of highly integrated and coordinated systems, machine intelligence sometimes overlooks the value of personalized needs and diversity. In practical applications, the diversity of user needs, data sources, and environments means that unified processing standards cannot be established. In contrast, human wisdom emphasizes diverse development routes that foster innovation and practical application. Therefore, machine intelligence should leverage systems engineering, advocate for diversity, and harness the advantages of different technologies and models through cross-domain integration.

Human intelligence evolves through diverse and heterogeneous means rather than uniform progression, with cognition valuing diversity to foster innovation and societal advancement. Diverse cultural backgrounds, ways of thinking, and individual experiences generate varying perspectives and methodologies, facilitating the identification of new problems and the creation of novel ideas. Embracing a heterogeneous development model promotes diversity, leading to numerous schools of thought and a flourishing of ideas. This approach mitigates the rigidity and limitations of homogenization, offering a more enriched and flexible developmental trajectory for society.

Emphasizing heterogeneous development over uniformity is crucial. Machine learning expert Michael I. Jordan claims that AI thrives through collective collaboration rather than individual efforts. For example, Jeff Dean and the Google Brain team have enhanced model performance and generalization by integrating diverse neural network architectures, algorithms, and data processing methods. This pluralistic approach demonstrates that collaboration among varied components fosters more robust intelligence than pursuing a single, regularized model. Similarly, DeepSeek,[9,10] a robust mixture-of-experts language model, achieves efficient inference and cost-effective training through such integration. As the Chinese proverb goes, three heads are better than one. By employing systems engineering for comprehensive integration, diverse technologies, models, and algorithms can leverage their strengths, ensuring specialized expertise and seamless coordination, ultimately achieving a synergistic effect.

CONCLUSION

Ray Kurzweil, an American computer scientist, predicts that machine intelligence will reach human levels by 2029 and surpass human wisdom by 2045. In contrast, Elon Musk has consistently warned that AI poses the greatest existential risk to humanity. These opposite views highlight the intense and concerning debate surrounding the advancement of machine intelligence toward true wisdom. This paper points to four major challenges that hinder machine intelligence: the neglect of silicon cognition, the lack of art, the trap of perfection, and the obsession with uniformity. We explore how leveraging the generative aspects of human wisdom, which are addressing foundational, methodological, and strategic dimensions, can enable machine intelligence to evolve to machine wisdom. The realization of machine wisdom has transformative potential for fields such as healthcare, education, and sustainability, enabling solutions to complex global challenges. Meanwhile, we also need to address ethical considerations and align AI development with human values to ensure societal well-being and a sustainable future.

REFERENCES

1. Fan, C., Zeng, L., Sun, Y. et al. (2020). Finding key players in complex networks through deep reinforcement learning. *Nat. Mach. Intell.* **2**:317–324.
2. Liu, S., Fan, C., Cheng, K. et al. (2024). Inductive meta-path learning for schema-complex heterogeneous information networks. *IEEE Trans. Pattern Anal. Mach. Intell.* **46**:10196–10209.
3. Xu, Y., Wang, F., An, Z. et al. (2023). Artificial intelligence for science—bridging data to wisdom. *Innovation* **4**:100525. DOI:https://doi.org/10.1016/j.xinn.2023.100525.
4. Felin, T. and Holweg, M. (2024). Theory is all you need: AI, human cognition, and causal reasoning. *Strateg. Sci.* **9**:346–371.
5. Xu, Y., Wang, F. and Zhang, T. (2024). Artificial intelligence is restructuring a new world. *Innovation* **5**:100725. DOI:https://doi.org/10.1016/j.xinn.2023.100525.
6. Strickland, E. and Ng, A. (2022). Unbiggen AI. The AI Pioneer Says it's Time for Smart-sized, "Data-centric" Solutions to Big Issues. *IEEE Spectrum* **9**.
7. Boden, M.A. (2009). Computer models of creativity. *AI Mag.* **30**:23–34.
8. Simon, H. (2020). Cognition: The Thinking and Intelligence behind Human Behavior (China: Renmin University of China Press).
9. Wu, Y., Zhang, Z., Wang, F., Xu, Y. and Huang, J. (2025). Towards more economical large-scale foundation models: no longer a game for the few. *The Innovation* **6**:100832. DOI:https://doi.org/10.1016/j.xinn.2025.100832.
10. Liu, A., Feng, B., Xue, B. et al. (2024). Deepseek-v3 technical report. preprint. *arXiv*. DOI:https://doi.org/10.48550/arXiv.2412.19437.

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DECLARATION OF INTERESTS

The authors declare no conflicts of interest.